



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to DEPT. OF SUGAR -SUBRAMANYA
SIVA CSM

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

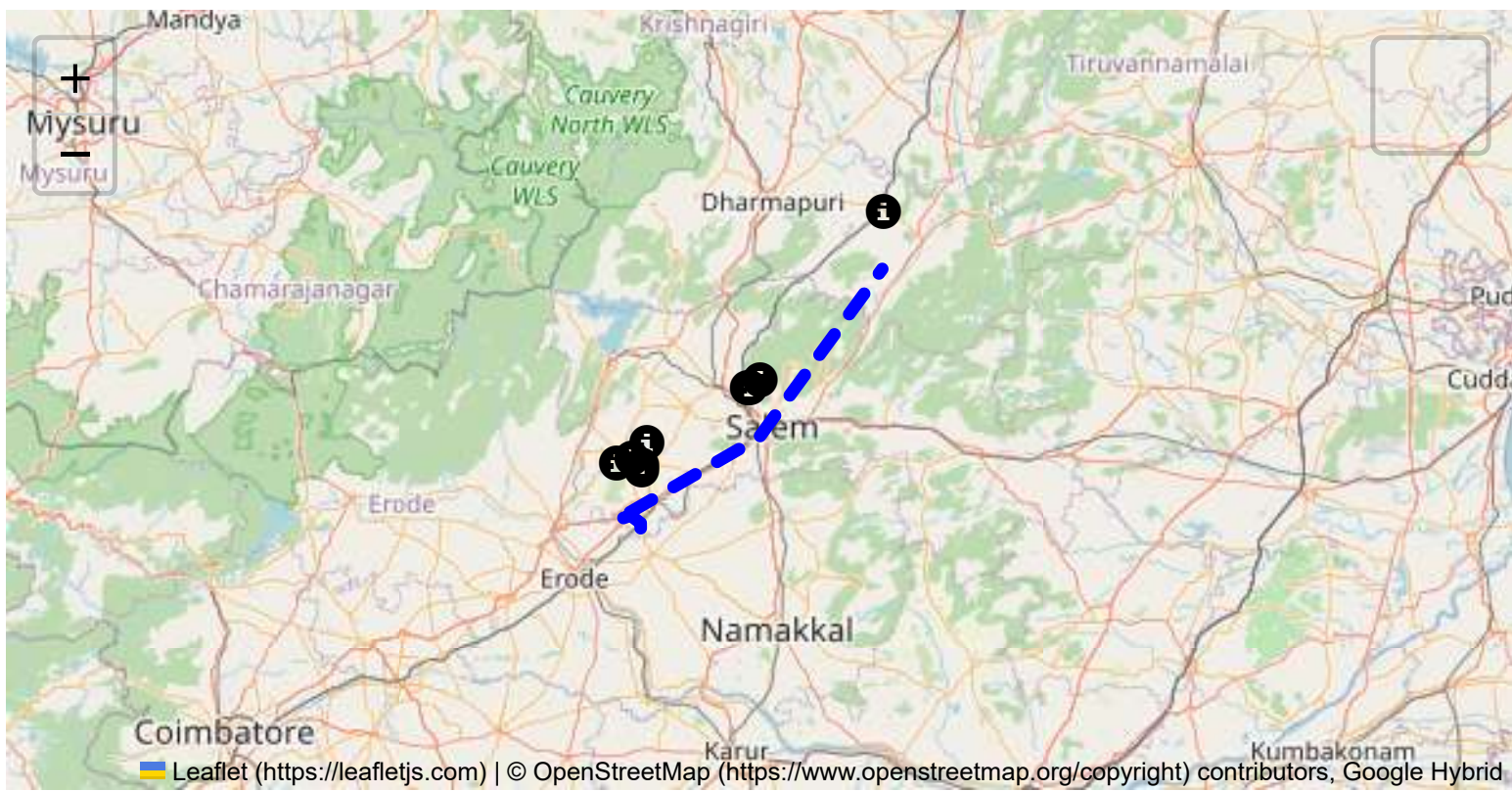
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

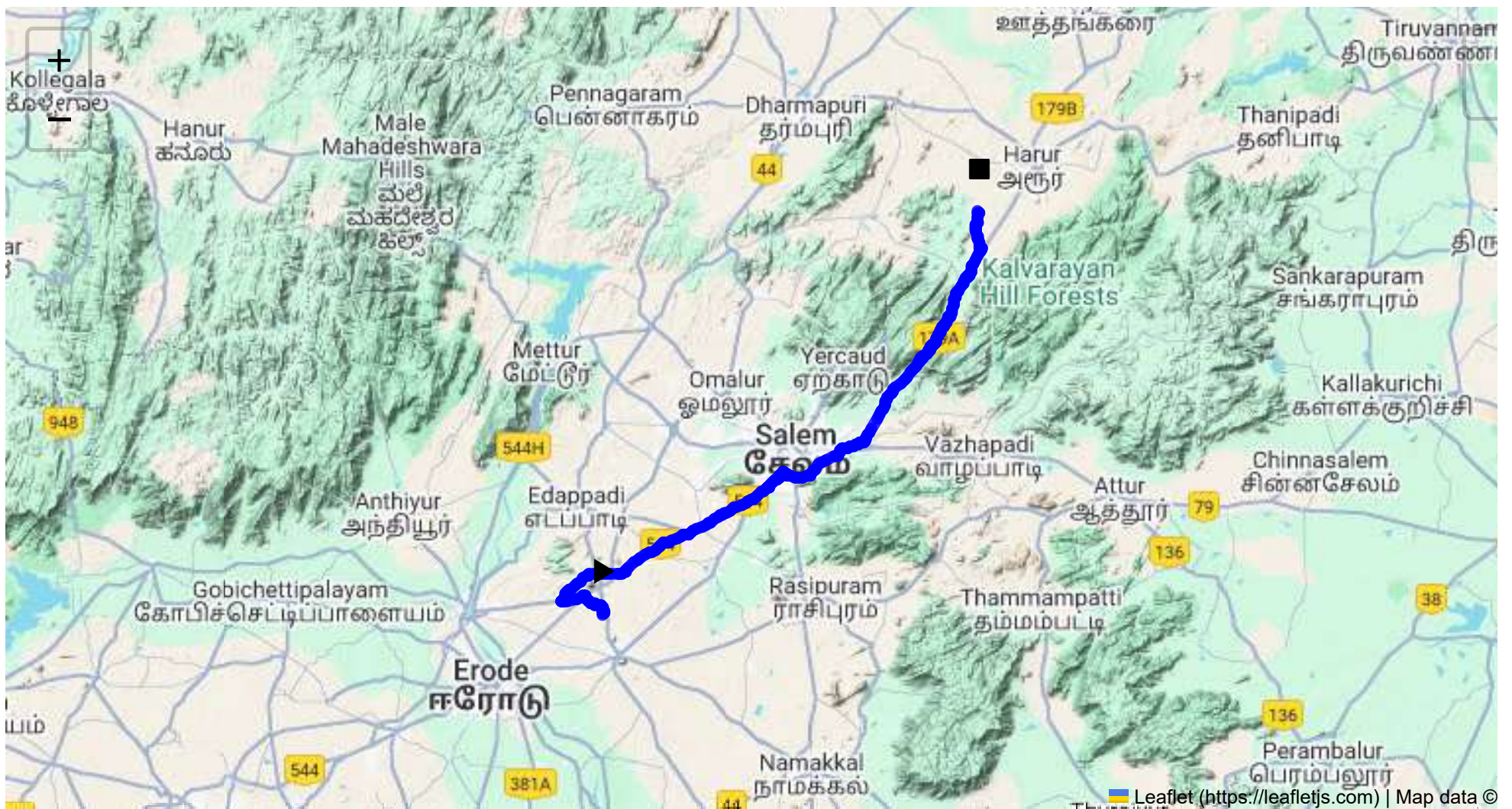
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 104.94 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.1 hours
Start: (11.4381, 77.8734)
End: (11.989621, 78.399601)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to S No 23/3A & 23/4, Gopalapuram, Alapuram spans approximately 104.94 kilometers. The path primarily includes rural roads and some sections of the national highway (NH544), entering the outskirts of Salem before reaching the final destination. The journey generally takes about 2.5 hours for trucks carrying hazardous materials.

1. Weather Conditions

Typical Weather: Tamil Nadu experiences a tropical climate. The route may face intense sun during peak summer months (March to June), while the monsoon season (July to November) can bring heavy rainfall and potentially slippery roads. Potential Hazards: Flash floods and reduced visibility during heavy rains and occasional cyclone threats during the monsoon season.

2. Traffic Patterns

- Peak Hours: Morning (8:00 – 10:00 AM) and evening (5:00 – 7:00 PM) are typically busy, especially around Salem and major junctions.
- Congestion Areas: The approach to Salem, particularly NH544, can see significant traffic buildup, especially near industrial and commercial areas.

3. Road Quality and Infrastructure

- The national highways are generally well-maintained, though some rural roads might have potholes or uneven surfaces.
- Narrow local roads can impact turning and maneuverability for large vehicles.

4. Emergency Routes

- In case of emergency, consider using lesser-traveled routes through nearby towns such as Tiruchengode, although this may extend travel time.
- Another alternative is using state highways that bypass major congestion points, particularly during peak hours or adverse weather.

5. Local Regulations on Hazardous Materials

- Adherence to government regulations on hazardous material transportation is crucial. Licenses and permits must be updated.
- Time-of-day restrictions for hazardous materials may apply in urban areas, restricting transit during peak human activity times.

6. Historical Incidents

- There have been a few documented incidents involving heavy vehicles, mostly due to driver fatigue or equipment failure. Safety measures must be strictly adhered to.

7. Environmental Considerations

- Proximity to rivers, especially during monsoons, requires caution to prevent contaminant spills.
- Several areas are agricultural zones; care must be taken to avoid environmental damage in case of spills.

8. Communication Coverage

- Generally, the route has good mobile and communication coverage, though some rural segments might experience weak signals.
- Major dead zones are minimal, but caution is advised to ensure communication devices are fully functional before traveling.

9. Emergency Response Times

- Urban areas like Salem offer relatively quick emergency services, but more remote parts of the route might delay response times, potentially up to 45 minutes.

10. Overall Summary

This route, while manageable, poses several risks associated with weather, road conditions, and traffic. Regular monitoring of weather forecasts and communication with local authorities for real-time traffic updates is recommended. Sticking to regulated transportation hours for hazardous materials is crucial to maintain safety and compliance. Robust emergency plans, including established contact points and alternative routes, should be in place to effectively mitigate risks.

Truck drivers should be well-briefed on these potential points of concern, and vehicles should be equipped for communication and safety. Regular maintenance checks are essential to prevent mechanical failures commonly associated with long-haul routes in challenging conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.53552, 77.95763	30 KM/Hr
11	Turn	High	11.61233, 78.10397	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.61216, 78.10423	15 KM/Hr
13	Turn	High	11.63300, 78.12746	15 KM/Hr
14	Turn	Medium	11.65825, 78.20469	30 KM/Hr
15	Turn	Medium	11.66366, 78.20649	30 KM/Hr
16	Turn	Medium	11.67129, 78.22978	30 KM/Hr
17	Turn	High	11.67120, 78.23136	15 KM/Hr
18	Turn	High	11.67182, 78.23141	15 KM/Hr
19	Turn	Medium	11.67422, 78.23881	30 KM/Hr
20	Turn	Medium	11.68017, 78.24250	30 KM/Hr
21	Turn	Medium	11.83514, 78.35305	30 KM/Hr
22	Turn	High	11.93905, 78.40417	15 KM/Hr
23	Turn	Medium	11.93960, 78.40395	30 KM/Hr
24	Turn	Medium	11.94039, 78.40200	30 KM/Hr
25	Turn	Medium	11.95570, 78.39975	30 KM/Hr
26	Turn	Medium	11.98751, 78.39991	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
19	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
20	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
21	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
22	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
23	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
24	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
25	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
28	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
29	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
26	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
27	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53552, 77.95763



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61233, 78.10397



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61216, 78.10423



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65825, 78.20469



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66366, 78.20649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67129, 78.22978



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67182, 78.23141



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68017, 78.24250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83514, 78.35305



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93905, 78.40417



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93960, 78.40395



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.94039, 78.40200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95570, 78.39975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.98751, 78.39991
